Introduction to image processing

Wigner Summer Camp

Data and Compute Intensive Sciences Research Group

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Imagine This...

- A self-driving car approaches a pedestrian crossing.
- A warehouse robot must avoid obstacles.
- A drone flies through a forest.

How do these machines know where objects are? They need to **see** and **understand** their environment.

Can Computers Really See?

A camera only captures millions of numbers.

$$\text{Image} = \text{Matrix of Pixels}$$

For a computer:

- A person is not a person.
- A road is not a road.
- A wall is not a wall.

Everything starts as numbers.

The Challenge

Humans instantly recognize:

- Roads
- Faces
- Buildings
- Obstacles

Computers must learn how to find useful information inside an image.
This is where **Computer Vision** begins.

What Is an Edge?

An edge is a place where brightness changes suddenly.

Examples:

- Border of a road
- Corner of a building
- Outline of a person
- Edge of a traffic sign

Edges often describe the shape of objects.

Why Is Edge Detection Important?

Without edge detection:

- Too much information
- Difficult to identify objects

With edge detection:

- Important structures become visible
- Less data to process
- Faster algorithms

Original Image vs Edge Image

Insert example images here:

Original Image $\rightarrow$ Edge Detection

Notice how most unnecessary information disappears while important structures remain.

Applications

Edge detection is used in:

- Self-driving cars
- Medical imaging
- Robotics
- Security systems
- Satellite image analysis
- Industrial quality control

What We Work On

Our projects include:

- Image processing
- Robot vision
- Object tracking
- Camera stabilization
- Neural networks
- Autonomous systems

All of these start with understanding images.